

```
#include <stdio.h>
```

```
#include <pthread.h>
```

```
#include <semaphore.h>
```

```
Sem_t empty, full;
```

```
Pthread_mutex_t mutex;
```

```
Int buffer = 0;
```

```
Void *producer(void *arg)
```

```
{
```

```
    Sem_wait(&empty);
```

```
    Pthread_mutex_lock(&mutex);
```

```
    Buffer = 1;
```

```
    Printf("Producer produced item %d\n", buffer);
```

```
    Pthread_mutex_unlock(&mutex);
```

```
    Sem_post(&full);
```

```
    Return NULL;
```

```
}
```

```
Void *consumer(void *arg)
```

```
{
```

```
    Sem_wait(&full);
```

```
Pthread_mutex_lock(&mutex);
```

```
Printf("Consumer consumed item %d\n", buffer);
```

```
Buffer = 0;
```

```
Pthread_mutex_unlock(&mutex);
```

```
Sem_post(&empty);
```

```
Return NULL;
```

```
}
```

```
Int main()
```

```
{
```

```
Pthread_t p, c;
```

```
Sem_init(&empty, 0, 1);
```

```
Sem_init(&full, 0, 0);
```

```
Pthread_mutex_init(&mutex, NULL);
```

```
Pthread_create(&p, NULL, producer, NULL);
```

```
Pthread_create(&c, NULL, consumer, NULL);
```

```
Pthread_join(p, NULL);
```

```
Pthread_join(c, NULL);
```

```
Sem_destroy(&empty);
```

```
Sem_destroy(&full);
```

```
Pthread_mutex_destroy(&mutex);
```

```
Return 0;
```

```
}
```